

To: Ricard Solé

Re: LiquidAI — proposed structural additions for ERC Synergy

Dear Ricard,

The vision document holds together. Four amendments would turn this into a Synergy with BCOM rather than an extension; the annex weaves them into your text (blue = additions). The conceptual logic follows.

Your proposal asks how collective intelligence emerges across substrates and what the morphospace of possible cognitions looks like. These are important questions — and they are also classical complex-systems questions. The BCOM contribution aims to sharpen the spine with three “Synergy” KT-driven questions that this program is uniquely positioned to answer.

1. When does a collective of agents become an agent? In KT terms, when does a higher-order Turing Pair (A_{coll}, W') come into existence — when does the collective acquire its own coherent ME (shared world model), OF (shared valence), and PE (shared counterfactual planning), running as a coupled loop at the collective timescale and over a representation space distinct from any individual member’s? This is what your complexity-drain thesis is reaching for, stated as a falsifiable claim about agency rather than a metaphor about complexity migration. Eusocial insects have crossed the threshold; markets are near-pure collective-OF with weak collective-PE; LLM populations are weak on all three but trending; the human brain has crossed all three completely. The same coordinates apply across substrates. The annex carries this as an explicit new §5.

2. What does the brain teach us about collective agency? The brain is the only paradigmatic, empirically accessible case of a collective-to-agent transition: 86 billion cells, themselves micro-agents, became a unified agent with structured experience. Computational neurophenomenology connects KT’s SEEN concept (Structured Experience Event — an episode of compressive-model execution) to neural dynamics; computational psychiatry shows what pathology of a cellular collective looks like from the inside. This is not a clinical appendix bolted onto a complex-systems project — it is the worked example that makes the project’s claims about collective agency concrete, testable, and normatively rich. The annex reframes WP5 around this: *The Brain as Paradigmatic Collective Agent — Computational Neurophenomenology, Mental Health, and the Diagnosis of Collective Pathology*.

3. Which realizations are flourishing vs. pathologic? The same module-level vocabulary that diagnoses individual mental disorders — ME fragmentation, OF capture, PE collapse, pathological SEENs — diagnoses collective pathology at every scale: epistemic crisis, preference manipulation, institutional decay, collective rumination. The brain teaches us what pathology looks like at the cell-collective scale; the project applies that vocabulary to human societies, AI agent populations, and hybrid ecologies. This is the project’s normative axis and its societal urgency, and it is what makes WP3 and WP4 genuinely BCOM-engaged rather than commentary on Solé-led work.

4. Agent Souper — a platform we should build. The proposal currently says we will simulate populations but commits to no substrate. WP6 operationalizes Class IV of WP0056 (our joint working paper) as an open, KT-architected multi-agent simulator on which all the

above quantities — and the detectors for the collective-agency transition itself — are first-class observables. Open source from day one, AGPL-3.0, MCP-compatible. This turns the project from “we will simulate” into “we will build the engine the program runs on.”

Architectural amendments in the annex. The three-link coupling $\text{Agents} \leftrightarrow \text{Agents} \leftrightarrow \text{Environment}$ with the channel–medium spectrum, and the equivalent world decomposition $W = \text{Self} \cup \text{OtherAgents} \cup \text{NonAgentEnv}$, are added to §1 (Figure 1). The KT vocabulary — Turing Pair, ME / OF / PE, memory as transmissible substrate, SEEN, Agent Soup, active vs. static execution — is introduced once at the end of §2 and referenced thereafter. A new §5 makes the collective-agency question explicit; the drain decomposition (Figure 2) gives the transition a numerical signature. WP5 is renamed and refocused on the brain as paradigmatic case; WP6 introduces Agent Souper.

Best,
Giulio